

Deep Learning

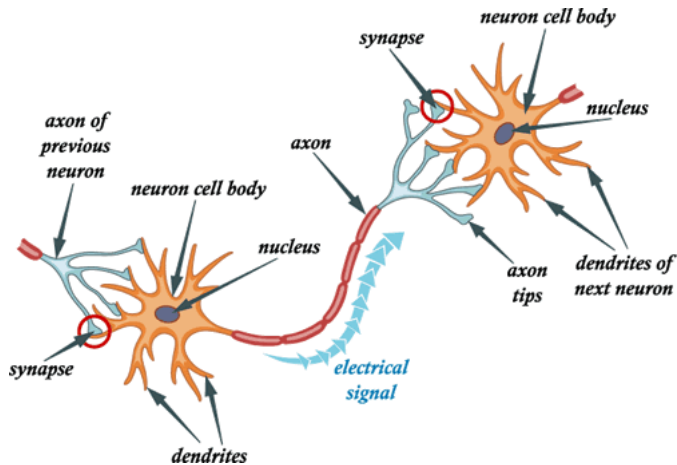
Vazgen Mikayelyan

May 8, 2021

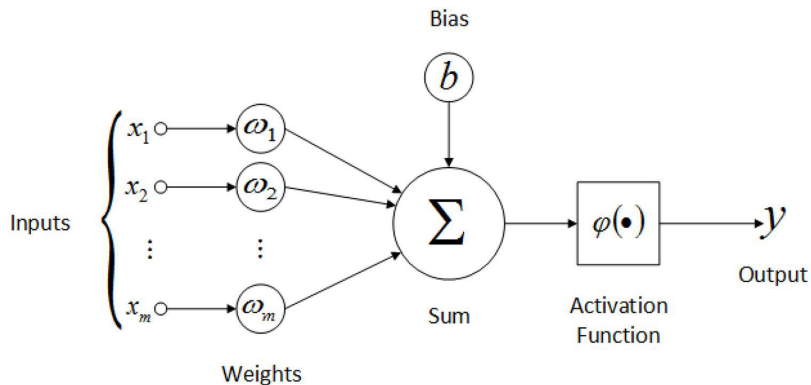


- 1 What is a neural network?
- 2 Gradient Descent
- 3 Linear and Logistic Regressions

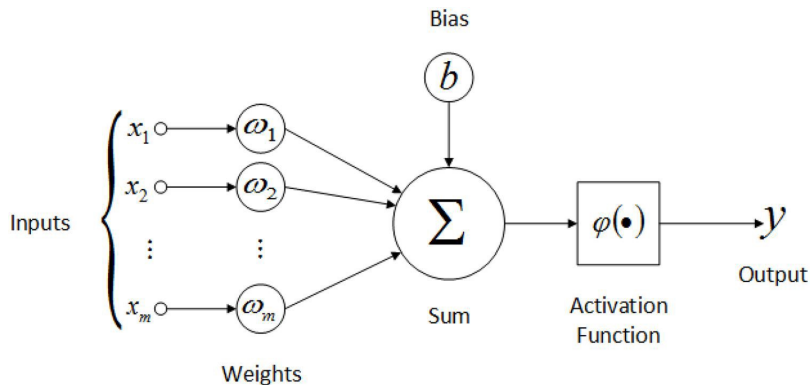
Human Brain



Artificial neuron



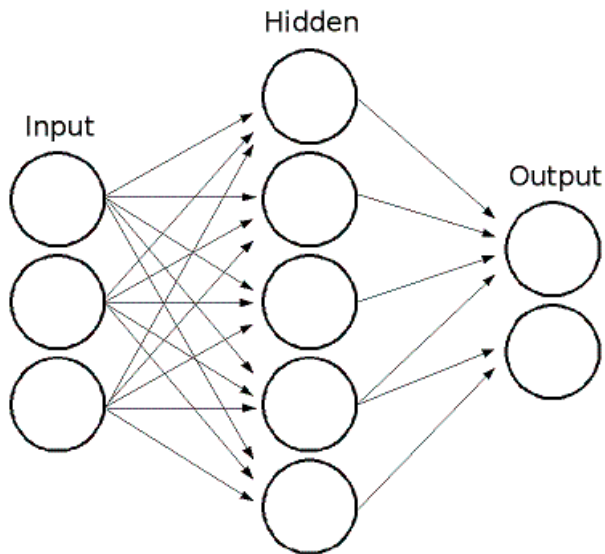
Artificial neuron



So artificial neuron is a function from $\mathbb{R}^m$ to $\mathbb{R}$, where m is the dimension of input:

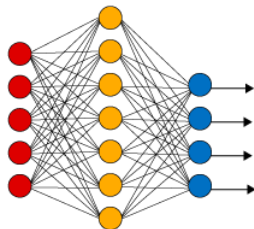
$$f_{w,b}(x_1, x_2, \dots, x_m) = \varphi(w_1x_1 + w_2x_2 + \dots + w_mx_m + b) = \varphi(w^T x + b)$$

Hidden Layer



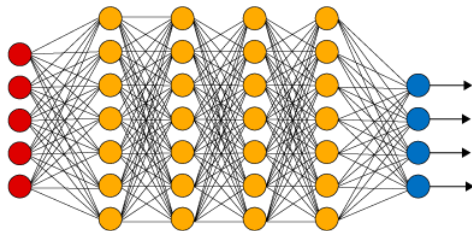
Deep neural network

Simple Neural Network



● Input Layer

Deep Learning Neural Network



● Hidden Layer

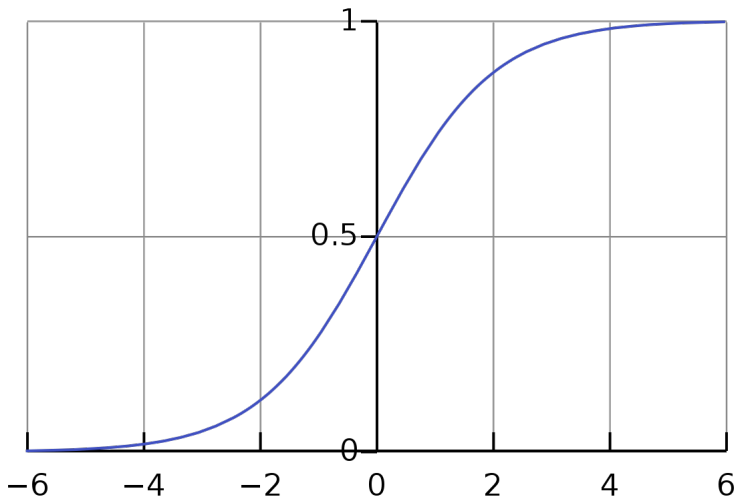
● Output Layer

Activation functions

1. Sigmoid: $\sigma(x) = \frac{1}{1 + e^{-x}}$

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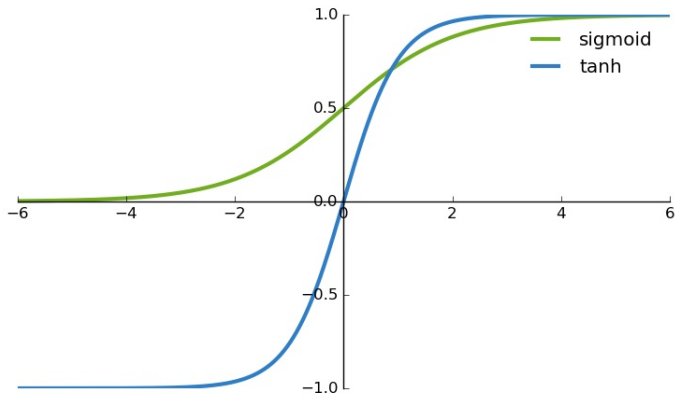


Activation functions

2. Tanh: $\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$

Activation functions

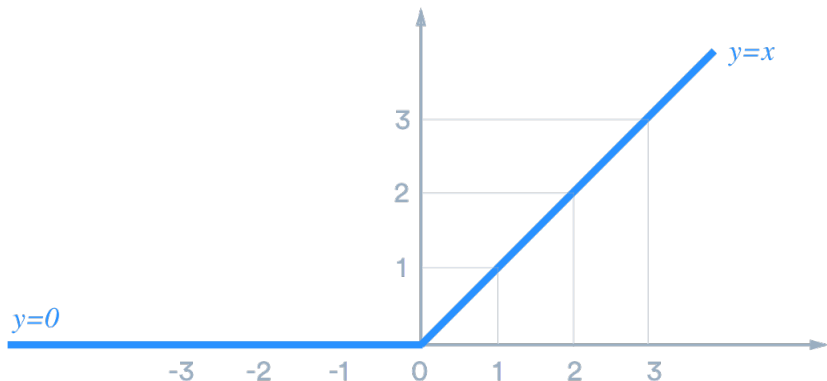
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3. Rectified linear unit: $\text{ReLU}(x) = \max(0, x)$

Activation functions

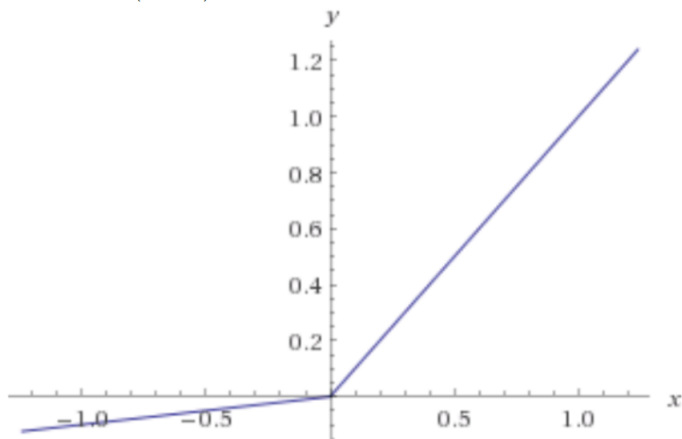
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4. Leaky ReLU: $LR(x) = \begin{cases} 0.01x, & \text{for } x < 0 \\ x, & \text{for } x \geq 0 \end{cases}$

Activation functions

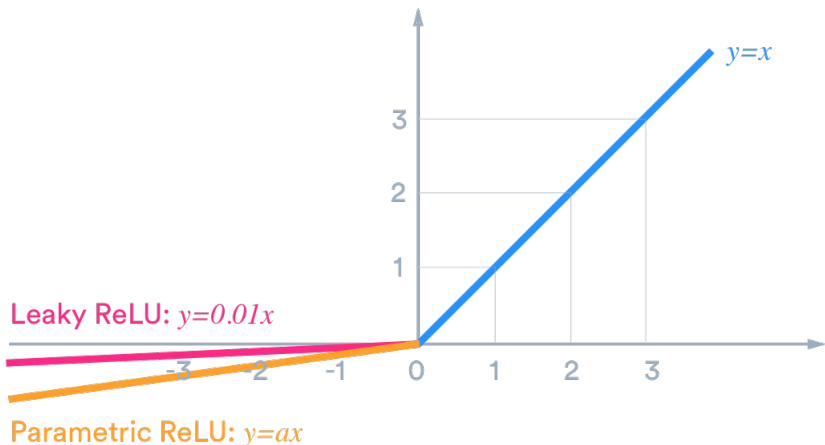
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5. Parametric ReLU: $PR(x) = \begin{cases} ax, & \text{for } x < 0 \\ x, & \text{for } x \geq 0 \end{cases}$

Activation functions

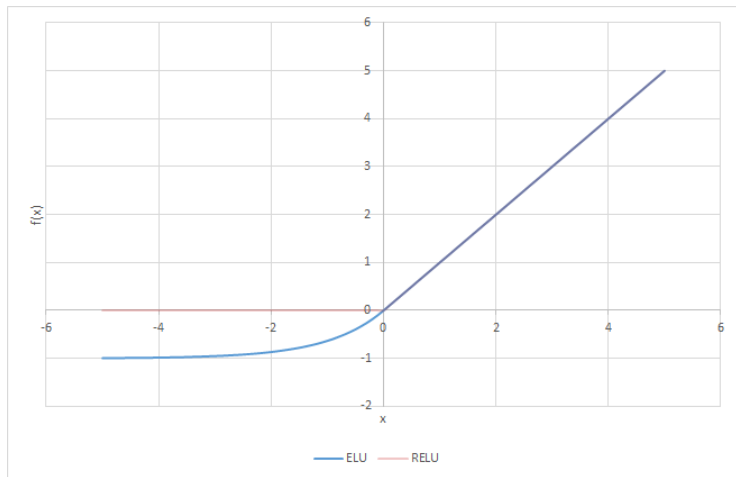
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6. Exponential linear unit: $ELU(x) = \begin{cases} a(e^x - 1), & \text{for } x < 0 \\ x, & \text{for } x \geq 0 \end{cases}$

Activation functions

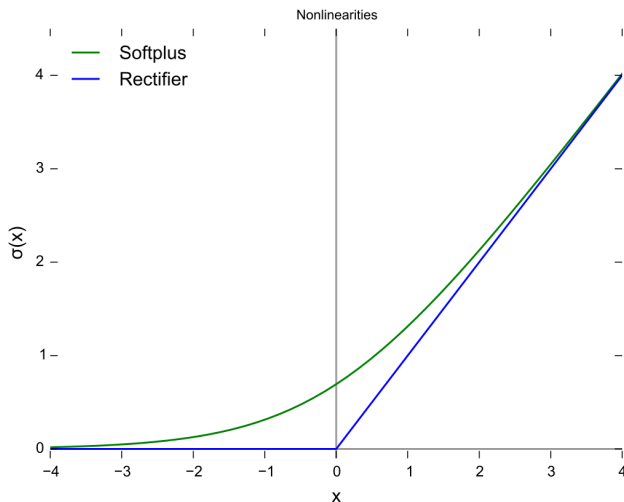
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Activation functions

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8. Softmax: $S(x_1, x_2, \dots, x_n) = \left(\frac{e^{x_1}}{\sum_{i=1}^n e^{x_i}}, \frac{e^{x_2}}{\sum_{i=1}^n e^{x_i}}, \dots, \frac{e^{x_n}}{\sum_{i=1}^n e^{x_i}} \right)$

- 1 Why do we need activation functions?

- ① Why do we need activation functions?
- ② How should we define activation functions, for a layer or for a neuron?

Outline

- 1 What is a neural network?
- 2 Gradient Descent**
- 3 Linear and Logistic Regressions

Gradient Descent

Let $f : \mathbb{R}^k \rightarrow \mathbb{R}$ be a convex function and we want to find its global minimum.

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$$x_{n+1} = x_n - \alpha \nabla f(x_n)$$

and $x_0 \in \mathbb{R}^k$ is a arbitrary point.

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Linear Regression

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Our aim is to find parameters $b, w^1, w^2, \dots, w^k$ such that

$$f(x_i) \approx y_i, i = 1, \dots, n.$$

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$$f(x_i) \approx y_i, i = 1, \dots, n.$$

We choose L^2 distance as our loss function:

$$\frac{1}{n} \sum_{l=1}^n (f(x_l) - y_l)^2.$$

- 1 Should we minimize the loss function using gradient descent?

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- 2 Can you represent this model as a neural network?

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We choose cross entropy distance as our loss function:

$$\frac{1}{n} \sum_{l=1}^n (-y_l \log f(x_l) - (1 - y_l) \log (1 - f(x_l))).$$